PADHAKU
Mathematics · Class 10 Mathematics



CHAPTER 7

Coordinate Geometry

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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❓ In-text questions (7.1 – 7.2)

? **Plot A(4, 8), B(3, 9), C(3, 8), D(1, 6), E(1, 5), F(3, 3), G(6, 3), H(8, 5), I(8, 6), J(6, 8), K(6, 9), L(5, 8) and join them as directed; join the triangles PQR, XYZ, STU and the lines from S and U. What picture have you got?**

💡 **SAMPLE ANSWER**

Plot the points on graph paper and join them in the given order.

The outline A → B → ... → L → A is a face with two pointed ears (at B and K).

Triangles PQR and XYZ are the two eyes, triangle STU is the nose, and the lines from S and U to the sides are the whiskers.

The picture is the face of a cat. (Check with your drawing.)

? **(Fig. 7.4) For P(6, 4) and Q(−5, −3), why is PT = 11 units and QT = 7 units?**

✔ **ANSWER**

PT is horizontal: it goes from x = −5 (line QS) to x = 6 (point P).

PT = 6 − (−5) = **11 units**.

QT is vertical: from y = −3 (point Q) up to y = 4 (level of P).

QT = 4 − (−3) = **7 units**.

So PQ = $\sqrt{(11^2 + 7^2)} = \sqrt{(121 + 49)} = \sqrt{170}$ units.

? **Remark 2: Why can we also write PQ = $\sqrt{[(x_1 - x_2)^2 + (y_1 - y_2)^2]}$?**

✔ ANSWER

$(x_1 - x_2) = -(x_2 - x_1)$, and squaring removes the minus sign: $(x_1 - x_2)^2 = (x_2 - x_1)^2$.

Similarly $(y_1 - y_2)^2 = (y_2 - y_1)^2$.

So both forms of the distance formula give the same value.

📝 Exercise 7.1

1 (i) **Find the distance between the points (2, 3) and (4, 1).**

✔ ANSWER

Distance = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

$= \sqrt{[(4 - 2)^2 + (1 - 3)^2]} = \sqrt{(4 + 4)} = \sqrt{8}$

$= 2\sqrt{2}$ units

1 (ii) **Find the distance between the points (–5, 7) and (–1, 3).**

✔ ANSWER

$= \sqrt{[(-1 + 5)^2 + (3 - 7)^2]} = \sqrt{(16 + 16)} = \sqrt{32}$

$= 4\sqrt{2}$ units

1 (iii) **Find the distance between the points (a, b) and (–a, –b).**

✔ ANSWER

$= \sqrt{[(-a - a)^2 + (-b - b)^2]} = \sqrt{(4a^2 + 4b^2)}$

$= 2\sqrt{(a^2 + b^2)}$ units

2 **Find the distance between the points (0, 0) and (36, 15). Can you now find the distance between the two towns A and B discussed in Section 7.2?**

✔ ANSWER

Distance $= \sqrt{[(36 - 0)^2 + (15 - 0)^2]} = \sqrt{(1296 + 225)} = \sqrt{1521} =$ **39 units**

Town B is 36 km east and 15 km north of town A, i.e. A(0, 0) and B(36, 15) with 1 unit = 1 km.

So the distance between the towns A and B is 39 km.

3 **Determine if the points (1, 5), (2, 3) and (–2, –11) are collinear.**

✔ ANSWER

Let A(1, 5), B(2, 3), C(–2, –11).

$AB = \sqrt{[(2 - 1)^2 + (3 - 5)^2]} = \sqrt{(1 + 4)} = \sqrt{5} \approx 2.24$

$BC = \sqrt{[(-2 - 2)^2 + (-11 - 3)^2]} = \sqrt{(16 + 196)} = \sqrt{212} \approx 14.56$

AC = $\sqrt{[(-2-1)^2 + (-11-5)^2]} = \sqrt{(9+256)} = \sqrt{265} \approx 16.28$

AB + BC ≈ 16.80 ≠ AC; no pair of distances adds up to the third.

So the points are not collinear.

4 **Check whether (5, −2), (6, 4) and (7, −2) are the vertices of an isosceles triangle.**

✔ ANSWER

Let A(5, −2), B(6, 4), C(7, −2).

AB = $\sqrt{[(6-5)^2 + (4+2)^2]} = \sqrt{(1+36)} = \sqrt{37}$

BC = $\sqrt{[(7-6)^2 + (-2-4)^2]} = \sqrt{(1+36)} = \sqrt{37}$

AC = $\sqrt{[(7-5)^2 + (-2+2)^2]} = \sqrt{4} = 2$

AB = BC, and AB + BC > AC, so a triangle is formed with two equal sides.

Yes, ABC is an isosceles triangle.

5 **In a classroom, 4 friends are seated at A, B, C and D (Fig. 7.8). Champa says ABCD is a square; Chameli disagrees. Using the distance formula, find who is correct.**

✔ ANSWER

From the figure: A(3, 4), B(6, 7), C(9, 4), D(6, 1).

AB = $\sqrt{[(6-3)^2 + (7-4)^2]} = \sqrt{(9+9)} = \sqrt{18} = 3\sqrt{2}$

BC = $\sqrt{[(9-6)^2 + (4-7)^2]} = \sqrt{(9+9)} = 3\sqrt{2}$

CD = $\sqrt{[(6-9)^2 + (1-4)^2]} = \sqrt{(9+9)} = 3\sqrt{2}$

DA = $\sqrt{[(3-6)^2 + (4-1)^2]} = \sqrt{(9+9)} = 3\sqrt{2}$

Diagonals: AC = $\sqrt{[(9-3)^2 + (4-4)^2]} = \sqrt{36} = 6$; BD = $\sqrt{[(6-6)^2 + (1-7)^2]} = \sqrt{36} = 6$

All four sides are equal and both diagonals are equal, so ABCD is a square.

Champa is correct.

6 (i) **Name the type of quadrilateral formed, if any, by the points (–1, –2), (1, 0), (–1, 2), (–3, 0), and give reasons.**

✔ ANSWER

Let A(–1, –2), B(1, 0), C(–1, 2), D(–3, 0).

AB = $\sqrt{[(1+1)^2 + (0+2)^2]} = \sqrt{(4+4)} = 2\sqrt{2}$

BC = $\sqrt{[(-1-1)^2 + (2-0)^2]} = \sqrt{(4+4)} = 2\sqrt{2}$

CD = $\sqrt{[(-3+1)^2 + (0-2)^2]} = \sqrt{(4+4)} = 2\sqrt{2}$

DA = $\sqrt{[(-1+3)^2 + (-2-0)^2]} = \sqrt{(4+4)} = 2\sqrt{2}$

AC = $\sqrt{[(-1+1)^2 + (2+2)^2]} = \sqrt{16} = 4$; BD = $\sqrt{[(-3-1)^2 + (0-0)^2]} = \sqrt{16} = 4$

All sides equal and diagonals equal.

ABCD is a square.

6 (ii) **Name the type of quadrilateral formed, if any, by the points (–3, 5), (3, 1), (0, 3), (–1, –4), and give reasons.**

✔ ANSWER

Let A(–3, 5), B(3, 1), C(0, 3), D(–1, –4).

AB = $\sqrt{[(3+3)^2 + (1-5)^2]} = \sqrt{(36+16)} = \sqrt{52} = 2\sqrt{13}$

BC = $\sqrt{[(0-3)^2 + (3-1)^2]} = \sqrt{(9+4)} = \sqrt{13}$

AC = $\sqrt{[(0+3)^2 + (3-5)^2]} = \sqrt{(9+4)} = \sqrt{13}$

AC + CB = $\sqrt{13} + \sqrt{13} = 2\sqrt{13}$ = AB, so A, C and B are collinear (C lies on AB).

Three of the four points lie on one line, so they cannot form a quadrilateral.

No quadrilateral is formed.

6 (iii) **Name the type of quadrilateral formed, if any, by the points (4, 5), (7, 6), (4, 3), (1, 2), and give reasons.**

✔ ANSWER

Let A(4, 5), B(7, 6), C(4, 3), D(1, 2).

AB = $\sqrt{[(7-4)^2 + (6-5)^2]} = \sqrt{(9+1)} = \sqrt{10}$

BC = $\sqrt{[(4-7)^2 + (3-6)^2]} = \sqrt{(9+9)} = \sqrt{18}$

CD = $\sqrt{[(1-4)^2 + (2-3)^2]} = \sqrt{(9+1)} = \sqrt{10}$

DA = $\sqrt{[(4-1)^2 + (5-2)^2]} = \sqrt{(9+9)} = \sqrt{18}$

AC = $\sqrt{[(4-4)^2 + (3-5)^2]} = \sqrt{4} = 2$; BD = $\sqrt{[(1-7)^2 + (2-6)^2]} = \sqrt{(36+16)} = \sqrt{52}$

Opposite sides are equal (AB = CD, BC = DA) but the diagonals are not equal.

ABCD is a parallelogram (not a rectangle).

7 **Find the point on the x-axis which is equidistant from (2, –5) and (–2, 9).**

✔ ANSWER

A point on the x-axis is P(x, 0). Let A(2, –5), B(–2, 9).

PA = PB ⇒ $PA^2 = PB^2$

$(x - 2)^2 + (0 + 5)^2 = (x + 2)^2 + (0 - 9)^2$

$x^2 - 4x + 4 + 25 = x^2 + 4x + 4 + 81$

$-4x + 29 = 4x + 85 \Rightarrow -8x = 56 \Rightarrow x = -7$

The point is (–7, 0).

8 **Find the values of y for which the distance between the points P(2, –3) and Q(10, y) is 10 units.**

✔ ANSWER

$PQ = 10 \Rightarrow PQ^2 = 100$

$(10 - 2)^2 + (y + 3)^2 = 100$

$64 + (y + 3)^2 = 100 \Rightarrow (y + 3)^2 = 36$

$y + 3 = 6$ or $y + 3 = -6$

y = 3 or y = –9

9 **If Q(0, 1) is equidistant from P(5, –3) and R(x, 6), find the values of x. Also find the distances QR and PR.**

✔ ANSWER

$QP = QR \Rightarrow QP^2 = QR^2$

$(5 - 0)^2 + (-3 - 1)^2 = (x - 0)^2 + (6 - 1)^2$

$25 + 16 = x^2 + 25 \Rightarrow x^2 = 16 \Rightarrow$ **x = ± 4**

$QR = \sqrt{(x^2 + 25)} = \sqrt{(16 + 25)} =$ **√41 units**

When x = 4, R(4, 6): $PR = \sqrt{[(4 - 5)^2 + (6 + 3)^2]} = \sqrt{(1 + 81)} =$ **√82 units**

When x = –4, R(–4, 6): $PR = \sqrt{[(-4 - 5)^2 + (6 + 3)^2]} = \sqrt{(81 + 81)} = \sqrt{162} =$ **9√2 units**

10 **Find a relation between x and y such that the point (x, y) is equidistant from the points (3, 6) and (−3, 4).**

✔ ANSWER

Let P(x, y), A(3, 6), B(−3, 4). PA = PB ⇒ $PA^2 = PB^2$

$(x - 3)^2 + (y - 6)^2 = (x + 3)^2 + (y - 4)^2$

$x^2 - 6x + 9 + y^2 - 12y + 36 = x^2 + 6x + 9 + y^2 - 8y + 16$

$-6x - 12y + 45 = 6x - 8y + 25$

$-12x - 4y + 20 = 0$ ⇒ divide by −4:

$3x + y - 5 = 0$

Exercise 7.2

1 **Find the coordinates of the point which divides the join of (−1, 7) and (4, −3) in the ratio 2 : 3.**

✔ ANSWER

Section formula: $x = (m_1x_2 + m_2x_1)/(m_1 + m_2)$, $y = (m_1y_2 + m_2y_1)/(m_1 + m_2)$, with $m_1 : m_2 = 2 : 3$.

$x = [2 \times 4 + 3 \times (-1)]/(2 + 3) = (8 - 3)/5 = 1$

$y = [2 \times (-3) + 3 \times 7]/5 = (-6 + 21)/5 = 3$

The point is (1, 3).

2 **Find the coordinates of the points of trisection of the line segment joining (4, –1) and (–2, –3).**

✔ ANSWER

Let A(4, –1), B(–2, –3) and P, Q be the points of trisection: AP = PQ = QB.

P divides AB in the ratio 1 : 2:

$x = [1 \times (-2) + 2 \times 4]/3 = 6/3 = 2$; $y = [1 \times (-3) + 2 \times (-1)]/3 = -5/3$

Q divides AB in the ratio 2 : 1:

$x = [2 \times (-2) + 1 \times 4]/3 = 0$; $y = [2 \times (-3) + 1 \times (-1)]/3 = -7/3$

The points of trisection are (2, –5/3) and (0, –7/3).

3 **In the rectangular school ground ABCD (Fig. 7.12), lines are drawn 1 m apart and 100 flower pots are placed 1 m apart along AD. Niharika runs 1/4 of the distance AD on the 2nd line and posts a green flag. Preet runs 1/5 of AD on the eighth line and posts a red flag. What is the distance between the flags? Where should Rashmi post a blue flag exactly halfway between the two flags?**

✔ ANSWER

Take A as origin, the lines as x (columns) and AD as the y-direction. AD = 100 m.

Niharika: 2nd line, 1/4 × 100 = 25 m ⇒ green flag at G(2, 25).

Preet: 8th line, 1/5 × 100 = 20 m ⇒ red flag at R(8, 20).

Distance GR = √[(8 − 2)² + (20 − 25)²] = √(36 + 25) = **√61 m**

Mid-point of GR = ((2 + 8)/2, (25 + 20)/2) = (5, 22.5)

Rashmi should post the blue flag on the 5th line, at a distance of 22.5 m from A (along the line).

4 **Find the ratio in which the line segment joining the points (–3, 10) and (6, –8) is divided by (–1, 6).**

✔ ANSWER

Let (–1, 6) divide the join of A(–3, 10) and B(6, –8) in the ratio k : 1.

x-coordinate: (6k − 3)/(k + 1) = −1

6k − 3 = −k − 1 ⇒ 7k = 2 ⇒ k = 2/7

Check y: [(−8)(2/7) + 10]/(2/7 + 1) = (54/7)/(9/7) = 6 ✓

The ratio is 2 : 7.

5 **Find the ratio in which the line segment joining A(1, –5) and B(–4, 5) is divided by the x-axis. Also find the coordinates of the point of division.**

✔ ANSWER

Let the x-axis divide AB in the ratio k : 1 at P.
On the x-axis, y = 0.

$y = (5k - 5)/(k + 1) = 0 \Rightarrow 5k = 5 \Rightarrow k = 1$

The ratio is 1 : 1 (P is the mid-point).

$x = (-4 \times 1 + 1 \times 1)/(1 + 1) = -3/2$

The point of division is (–3/2, 0).

6 **If (1, 2), (4, y), (x, 6) and (3, 5) are the vertices of a parallelogram taken in order, find x and y.**

✔ ANSWER

Let A(1, 2), B(4, y), C(x, 6), D(3, 5). Diagonals of a parallelogram bisect each other, so mid-point of AC = mid-point of BD.

$((1 + x)/2, (2 + 6)/2) = ((4 + 3)/2, (y + 5)/2)$

$(1 + x)/2 = 7/2 \Rightarrow x = 6$

$8/2 = (y + 5)/2 \Rightarrow y + 5 = 8 \Rightarrow y = 3$

x = 6, y = 3

7 **Find the coordinates of a point A, where AB is the diameter of a circle whose centre is (2, –3) and B is (1, 4).**

✔ ANSWER

The centre is the mid-point of the diameter AB. Let A(x, y).

$(x + 1)/2 = 2 \Rightarrow x + 1 = 4 \Rightarrow x = 3$

$(y + 4)/2 = -3 \Rightarrow y + 4 = -6 \Rightarrow y = -10$

A = (3, –10)

8 **If A and B are (−2, −2) and (2, −4) respectively, find the coordinates of P such that AP = 3/7 AB and P lies on the line segment AB.**

✔ ANSWER

AP = (3/7) AB ⇒ PB = AB − AP = (4/7) AB ⇒ AP : PB = 3 : 4.

x = [3 × 2 + 4 × (−2)]/(3 + 4) = (6 − 8)/7 = −2/7

y = [3 × (−4) + 4 × (−2)]/7 = (−12 − 8)/7 = −20/7

P = (−2/7, −20/7)

9 **Find the coordinates of the points which divide the line segment joining A(−2, 2) and B(2, 8) into four equal parts.**

✔ ANSWER

Let P, Q, R divide AB into four equal parts; Q is the mid-point of AB, P the mid-point of AQ, R the mid-point of QB.

Q = ((−2 + 2)/2, (2 + 8)/2) = (0, 5)

P = ((−2 + 0)/2, (2 + 5)/2) = (−1, 7/2)

R = ((0 + 2)/2, (5 + 8)/2) = (1, 13/2)

The points are (−1, 7/2), (0, 5) and (1, 13/2).

10 **Find the area of a rhombus if its vertices are (3, 0), (4, 5), (−1, 4) and (−2, −1) taken in order. [Hint: Area of a rhombus = ½ (product of its diagonals)]**

✔ ANSWER

Let A(3, 0), B(4, 5), C(−1, 4), D(−2, −1).

Diagonal AC = $\sqrt{[(-1-3)^2 + (4-0)^2]} = \sqrt{(16+16)} = \sqrt{32} = 4\sqrt{2}$

Diagonal BD = $\sqrt{[(-2-4)^2 + (-1-5)^2]} = \sqrt{(36+36)} = \sqrt{72} = 6\sqrt{2}$

Area = ½ × AC × BD = ½ × $4\sqrt{2}$ × $6\sqrt{2}$ = ½ × 48

Area = 24 square units

🎉 **Chapter 7 done! Revise the answers once more tomorrow.**